Computational Finance and Financial Technology

	Normative 4-year Degree
Major	Computational Finance and Financial Technology 計算金融及金融科技
Degree / Award Title	Bachelor of Science in Computational Finance and Financial Technology 理學士(計算金融及金融科技)
Offering Academic Unit	This Major is jointly offered by: College of Business Department of Economics and Finance Department of Information Systems Department of Decision Analytics and Operations
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	124 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2402 English for Business Communication	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology	12

	Visit the GE website for more course information.	
College-specified Courses *(for ComFin Stream)*	CB2240 Introduction to Business Programming in Python	3
	CS3334 Data Structures	3
	MA3525 Elementary Numerical Methods	3
College-specified Courses *(for FinTech Stream)*	CB2240 Introduction to Business Programming in Python	3
	CS2312 Problem Solving and Programming	3
	MA2185 Discrete Mathematics	3

College / School Requirements (9 credit units) — Catalogue Term : Semester A 2025/26

Course Code	Course Title	Credit Units
CB2100	Introduction to Financial Accounting	3
CB2400	Microeconomics	3
CB3410	Financial Management	3

Major Requirements (72 credit units) — Catalogue Term : Semester A 2024/25

1. Core Courses (30 credit units)

Select **One** from the following courses:

Course Code	Course Title	Credit Units
MA1200	Calculus and Basic Linear Algebra I	3
MA1300	Enhanced Calculus and Linear Algebra I	3

Select **One** from the following courses:

Course Code	Course Title	Credit Units
MA1201	Calculus and Basic Linear Algebra II	3
MA1301	Enhanced Calculus and Linear Algebra II	3

Take **ALL** the below courses:

Course Code	Course Title	Credit Units
MA2510	Probability and Statistics	3
CS1102	Introduction to Computer Studies	3
CS2311	Computer Programming	3
CS3402	Database Systems	3
EF3320	Security Analysis and Portfolio Management	3
EF4313	Corporate Finance	3
EF4321	Derivatives and Risk Management	3
MS3252	Regression Analysis	3

a) Computational Finance Stream

i) Stream Core Courses (36 credit units)

Course Code	Course Title	Credit Units	Remarks
MA2001	Multi-variable Calculus and Linear Algebra	3	
MA3511	Ordinary Differential Equations	3	
CB2402	Macroeconomics	3	
EF3520	Stochastic Calculus for Finance	3	
EF4820	Derivatives Pricing I: Stock and FX	3	
EF4821	Derivatives Pricing II: Interest Rate and Credit Risk	3	
EF4822	Financial Econometrics	3	
EF4328	Asset Management	3	
MS2602	Statistical Inference	3	
MS3111	Data Analytics with Excel VBA	3	
MS3601	Optimization Methods	3	
CB4001	Honor Thesis	3	This course can be used to fulfil the internship/consultancy project/research project requirement.

ii) Stream Electives (6 credit units)

Students are required to take **TWO** elective courses from the list below

Course Code	Course Title	Credit Units
CB2300	Management	3
CB3043	Business Case Analysis and Communication	3
CS3391	Advanced Programming	3
CS4335	Design and Analysis of Algorithms	3
EF4312	Mergers and Acquisitions	3
EF4314	Corporate Valuation	3
EF4323	Trading Room Workshop	3
EF4327	Fixed Income Securities	3
EF4331	International Finance and Banking	3
EF4334	Regulation and Management of Financial Institutions	3
MS3106	Simulation	3
MS4212	Predictive Analytics and Forecasting	3
MS4224	Enterprise Data Mining	3

MS4252	Big Data Analytics	3
MA3514	Numerical Methods for Differential Equations	3
MA4542	Real Analysis	3

b) Financial Technology Stream

i) Stream Core Courses (36 credit units)

Course Code	Course Title	Credit Units	Remarks
CB2500	Information Management	3	
CS3334	Data Structures	3	
EF4314	Corporate Valuation	3	
IS3101	Blockchain and Digital Currency	3	
IS3240	Advanced Business Programming in Python	3	
IS3501	Cybersecurity for Business	3	
IS4837	Regulatory Compliance for Financial Services Technologies	3	
IS4335	Data Visualization	3	
IS4861	Machine Learning for Business	3	
IS4920	FinTech Capstone Project	6	This course can be used to fulfil the internship/consultancy project/research project requirement.
IS4940	Global Technology Entrepreneurship and Innovation	3	

ii) Stream Electives (6 credit units)
Students are required to take **TWO** elective courses from the list below.

Course Code	Course Title	Credit Units
CB2101	Introduction to Managerial Accounting	3
CB2201	Operations Management	3
CB2300	Management	3
CB2402	Macroeconomics	3
CB2601	Marketing	3
CB3043	Business Case Analysis and Communication	3
EF4312	Mergers and Acquisitions	3
EF4323	Trading Room Workshop	3
IS2502	Social Media and Social Networks	3
IS3230	Java Programming for Business	3
IS3430	Systems Analysis and Design	3

IS4032	Mobile Apps	3
IS4537	Information Systems Audit	3
IS4543	Risk Management and Information Systems Control	3
MKT3603	Consumer Behaviour	3
MGT2324	Introduction to Entrepreneurship	3

Suggested Study Plan

Suggested Study Plan (Cohort 2025 Normative 4-year Degree) - Computational Finance Stream

Suggested Study Plan (Cohort 2025 Normative 4-year Degree) - Financial Technology Stream